

2. The following results were obtained from an investigation into stretching a spring.

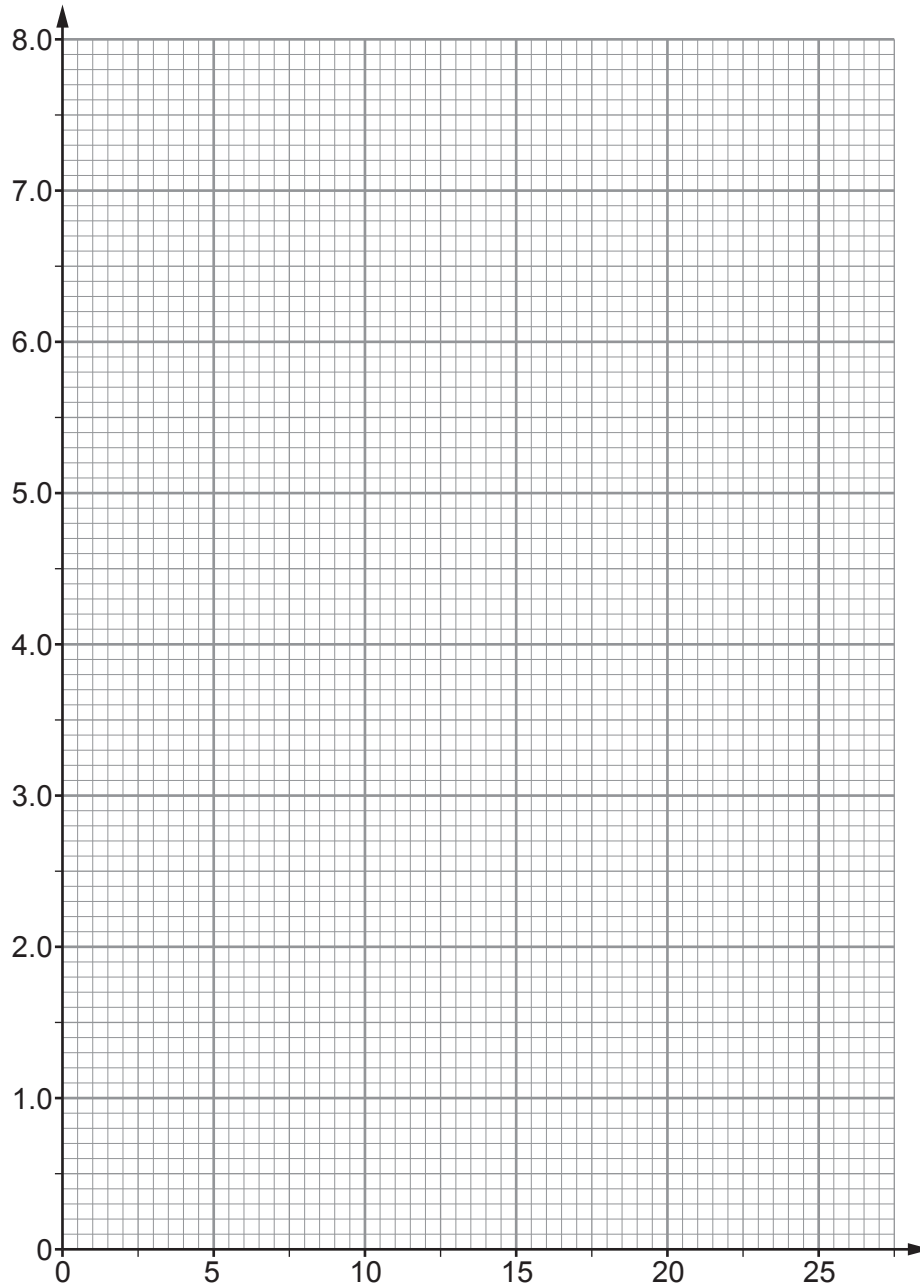
Force (N)	Length of spring (cm)
1.0	12.0
2.5	15.0
4.0	18.0
5.0	20.0
6.0	22.0
7.5	25.0



(a) Plot the data on the grid below and draw a suitable straight line.

[3]

Force (N)



Length of spring (cm)



(b) Use the graph to answer the following questions.

- (i) The unstretched length of the spring is its length when the force applied is zero.
State the unstretched length of the spring. [1]

Unstretched length = cm

- (ii) I. State the length of the spring produced by a force of 3.0 N. [1]

Length = cm

II. Use the equation:

extension = length produced by 3.0 N – unstretched length

to calculate the extension produced by a force of 3.0 N. [1]

Extension = cm

- (iii) Use your previous answer and the equation:

$$\text{spring constant} = \frac{\text{force}}{\text{extension}}$$

to calculate the spring constant of the spring for a force of 3.0 N. [2]

Spring constant = N/cm

- (iv) Complete the sentence by underlining the correct answer in the brackets. [1]

If the force on the spring is doubled to 6.0 N, then the spring constant
(**halves** / **stays the same** / **doubles**).

